

HILARY OMONDI OTIENO

Backend Engineer & Machine Learning Specialist

Kisumu, Kenya | +254 759 865 977 | hilaryomondi08@gmail.com

github.com/luckylittleman | <https://www.linkedin.com/in/hilary-omondi-867209273/> | hilaryomondi.netlify.app

PROFESSIONAL SUMMARY

Computer Science student with hands-on experience across full-stack development, machine learning, and network engineering. Skilled in building backend systems with Python/FastAPI and frontends with React, with production experience managing database architecture. Built predictive analytics models and a modular neural network framework from scratch, and optimized enterprise network infrastructure during an industrial attachment. Detail-oriented, fast learner, and comfortable working in Linux environments.

Core Competencies: Teamwork & Collaboration • Problem-Solving • Discipline & Time Management • Attention to Detail • Adaptability

TECHNICAL SKILLS

- Languages: Python, JavaScript, SQL, HTML/CSS
- Frameworks & Libraries: FastAPI, Django, Django REST Framework, React, NumPy, Pandas, Scikit-learn, Matplotlib
- Databases & Tools: PostgreSQL, SQLite, Git, GitHub
- Networking & Security: Router & Switch Configuration, Network Troubleshooting, WAN/LAN Design, SSH Hardening, Wireshark
- Systems: Linux (Ubuntu)

PROFESSIONAL EXPERIENCE

AnalystLab Africa — Remote Jul 2026 – Present

Machine Learning Intern

- Follow a disciplined, weekly project-based ML curriculum covering data preprocessing, exploratory data analysis, supervised learning, and ensemble methods, consistently meeting delivery deadlines.
- Built and evaluated classification and regression models, systematically comparing algorithm performance to troubleshoot and select optimal approaches.

Kalanet Communications — Kisumu, Kenya May 2025 – Aug 2025

Industrial Attaché, Network Engineering / IT Operations

- Monitored and maintained network infrastructure, sustaining consistent uptime for enterprise and residential clients.
- Diagnosed and resolved complex connectivity issues under tight response windows, applying structured troubleshooting to isolate root causes quickly.
- Partnered closely with senior network engineers on routing optimization, communicating clearly to document infrastructure changes for the wider team.

TECHNICAL PROJECTS

Student Dropout Predictor Python, Machine Learning, NumPy, Pandas

- Designed and built a machine learning system to identify at-risk university students in Kenya using academic and financial data.

NumPy-Net Python, NumPy, Machine Learning

- Architected a modular deep learning neural network framework built entirely from scratch using NumPy primitives.

Full-Stack E-Commerce Platform Django, Django REST Framework, React, JavaScript

- Built a full-stack e-commerce web application with user authentication, product listings, shopping cart state management, and order processing.

EDUCATION

Bachelor of Science in Computer Science Sept 2022 – Nov 2026 (Expected)

Maseno University, Main Campus

- Relevant Coursework: LAN Design & Implementation, Database Management Systems, Object-Oriented Programming, Data Structures & Algorithms, Software Engineering

Kenya Certificate of Secondary Education (KCSE) Jan 2018 – Apr 2022

Kisumu Boys High School

INTERESTS

Chess and strategic board games